Reinsuring Private Credit by Private Equity *

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Abstract

Over the last decade, U.S. life insurers have aggressively shifted reserves to offshore jurisdictions while simultaneously reallocating portfolios toward private credit. This paper finds that the two are linked through private equity reinsurers domiciled in Bermuda. Insurers benefit directly or indirectly from exposure to higher yielding private credit and the more competitive reinsurance market. Private equity with a large presence in Bermuda can access desired blocks of life insurer capital for their private credit investments as a reinsurer. Using comprehensive regulatory filings from 2011-2024, we exploit the staggered adoption of offshore reinsurance agreements, to estimate the causal impact of these arrangements on portfolio allocation. Two years following the execution of such a reinsurance agreement with a private-equity backed reinsurer, insurers increase their investments in NAIC 1 long-term bonds in private credit by 1.4 percentage points or a third of the median share of investments in private credit. In addition, these reinsurance and portfolio allocation decisions coincide with changes in the annuity product market. Two years following the execution of such a reinsurance agreement, insurers also increase their annual total annuity premiums (lump sum deposits) collected by around $0.82 billion.

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1 Introduction

During the past decade, the US life insurance industry has undergone significant structural changes. First, the use of life reinsurance, which helps with the management of more than $1 trillion in assets, has increased substantially (figure 1). Although the majority of reinsurance was historically provided domestically, offshore reinsurers - predominantly those domiciled in Bermuda - have now become the primary providers by some measures (figure 2). Second, following broader trends in the rapid growth of private credit, life insurers have emerged as major investors, accounting for around a third of all private credit investments in the US. Third, private equity (PE) firms have increasingly targeted life insurers and their large pools of capital, leading to many PE acquisitions of life insurers over the past decade.

In this paper, we examine why the life reinsurance market has experienced such a large shift toward offshore reinsurers. Our answer ties these trends in reinsurance to concurrent developments in private credit and PE together. Leveraging the staggered execution of reinsurance agreements in a difference-in-difference research design, we find that following a reinsurance agreement with a PE-backed reinsurer domiciled in Bermuda, life insurers significantly increased their investments in NAIC 1 designated long-term bonds in private credit.[1] These insurers also significantly increased their annuity sales, which make up the majority of reinsurance ceded.[2] In contrast, we find no such effects when the reinsurer is not backed by PE.

We interpret these results through the following mechanism. To support their growth, especially in interest-rate sensitive annuity business, life insurers have sought higher yielding investments, such as private credit. However, unilaterally investing in more illiquid and hard to value assets carries increased risks in servicing current policyholder claims. Reinsurance allows insurers to benefit from exposure to private credit while mitigating these underlying risks. On the reinsurer side, many PE funds and PE-backed reinsurers are currently domiciled in Bermuda due to tax advantages, an internationally recognized regulatory regime and a specialized reinsurance infrastructure. Private credit is closely linked to private equity: many private credit managers are part of firms active in PE, and many private credit borrowers are PE-sponsored companies. The International Monetary Fund (2024) reports that asset managers whose umbrella firm is also active in PE hold more than three-quarters of private credit assets, and that PE firms sponsor roughly 70

[1] NAIC designations, are regulatory investment-quality categories used in statutory reporting and risk-based-capital regulation for insurer-owned securities. The scale runs from NAIC 1 to NAIC 6, with lower numbers indicating lower investment risk. NAIC 1 is roughly comparable to the higher investment-grade ratings, spanning Moody's Aaa through A3 or S&P/Fitch AAA through A–; NAIC 2 corresponds roughly to Baa/BBB ratings; NAIC 3 to Ba/BB; NAIC 4 to B; NAIC 5 to Caa/CCC; and NAIC 6 to securities in or near default.

[2] Insurers are often referred to as 'cedants' who 'cede' insurance premiums and reserves to reinsurers who are often said to 'assume' insurance premiums and reserves from insurers.

percent of private credit deals. Life insurers are an ideal source of capital for private credit because their long-duration, stable liabilities allow them to hold less liquid assets to maturity, capture illiquidity premia, and match the cash flows of long-term annuity obligations. While PE firms could access life insurer capital by acquiring primary insurers directly, they may prefer to avoid taking on an insurer's entire block of insurance business and their associated liabilities (such as health lines of business, which have more liquid cashflow obligations). Reinsurance allows PE firms to access specific blocks of insurance and their associated assets and liabilities, utilizing their large balance sheets to absorb the risks from private credit investments. Thus, offshore life reinsurance provides primary life insurers exposure to higher-yielding assets such as private credit, while granting PE-backed reinsurers access to capital for their private credit investments.

To conduct our analysis, we combine life insurer annual statutory financial statements filed with the National Association of Insurance Commissioners (NAIC) to construct a comprehensive panel of life insurance companies from 2011-2024 and their investments and executed reinsurance agreements. To track an insurer's investments, we draw on Schedule D and Schedule BA, which collectively detail an insurer's long-term investment holdings. Because investments are not explicitly tagged as private credit in statuary filings (reflecting the lack of a universal definition of private credit), we draw on a broader set of investment characteristics to categorize investments as private credit. For reinsurance, we utilize Schedule S to capture data on all active reinsurance agreements. While these files provide reinsurer identities, NAIC regulatory filings do not provide a systematic way of identifying whether a Bermuda-domiciled reinsurer is PE-backed or not. To overcome this limitation, we hand-collect information to identify the PE-affilation of each Bermuda-domiciled reinsurer in our sample.

Importantly, our analysis is focused on reinsurance agreements where the assets supporting the reserves in the agreements are retained on the insurer's balance sheet. These include modified coinsurance ('modco') and funds withheld reinsurance agreements.[3] This differs from traditional coinsurance agreements where the assets are transferred to the reinsurer's balance sheet. Thus, for modco and funds withheld reinsurance agreements, we can study how the investments supporting the reinsurance agreement change. Since the majority of life reinsurance is modco or funds withheld, this is without loss of generality (figure 5). Note that while we do not directly observe the treaty-specific asset pool for each reinsurance agreement, our research design relies on changes in the insurer's overall asset portfolio, following the execution of the reinsurance agreement.

We begin the analysis by documenting important trends in life reinsurance, private credit and private equity. While life reinsurance, including offshore reinsurance is hardly new, we document two large shifts in reinsurance that were in part driven by regulatory changes. First, regulations

[3]Various transactions and accounting treatments ensure that the risk is still transferred to the reinsurer as discussed in section 2.

governing captive reinsurers were tightened in 2015.[4] Captive reinsurers are reinsurers owned and created for the purpose of reinsuring policies written by the insurer. This led to dramatic reductions in the usage of captive reinsurers (figure 4). Second, beginning in 2015, regulations surrounding collateral requirements for reinsurers were relaxed depending on the jurisdiction of the reinsurer. This likely contributed to the expansion of the Bermuda reinsurance market and the growth in reinsurance ceded to Bermuda. As a result, the average reinsurance agreement in the decade beginning 2015 looked quite different to the average reinsurance agreement prior. Accordingly, our difference-in-difference design focuses on reinsurance agreements executed from 2015 onward.

Turning to private credit, we document that almost all of life insurers' investments in private credit are in long-term, bond-like securities with a significant proportion being NAIC 1 designated (figure 7). Regarding PE ownerships, using our hand-cllected list of Bermuda domiciled PE-backed reinsurers and NAIC's list of PE-backed primary insurers, we reveal a stark asymmetry: the majority of private credit investments are still held by non-PE backed primary insurers, yet the majority of reserves ceded to Bermuda reinsurers are captured by PE-backed reinsurers. While considerable scrutiny has been applied to PE-owned life insurers, our descriptive facts suggest that the relative importance of PE ownership of reinsurers may also warrant the same, if not more scrutiny.

At the core of our empirical analysis is a staggered difference-in-difference research design that studies the impact of executing a post-2015 modco or funds withheld Bermuda reinsurance agreement on an insurer's private credit investments and annuity sales. The research design compares insurers who have executed such an agreement, against a control group of firms who have not yet done so, alongside those who never utilize offshore reinsurance. When the reinsurer is PE-backed we find strong growth in the share of NAIC 1 long-term bond private credit investments. Two years following the execution of the reinsurance agreement, insurers increase their share of investments in NAIC 1 private credit by about 1.40 percentage points, or just over a third of their median share of investments in private credit (in 2024). This contrasts with the result for non-PE reinsurers, where we find no changes to the insurer's investments following the execution of the reinsurance agreement. The increase in the share of NAIC 1 private credit is also persistent, growing to around 3 percentage points 5 years from execution of the reinsurance agreement. To contextualize the role of PE in life reinsurance versus primary insurance, we compare our results to existing estimates of direct PE acquisitions. While our estimates are smaller, we think the aggregate effects are similar, as our treated sample of insurers is much larger than the group of insurers who have been acquired by PE.

[4] Although some states had begun regulatory actions prior. For example, these issues were highlighted in a public report by the New York Department of Financial Services in 2013.

Furthermore, we find that the bulk of the effect comes from affiliated reinsurance agreements (when the insurer and reinsurer are related, including for example, through a parent company) rather than unaffiliated reinsurance agreements. For example, 2 years following the execution of an affiliated reinsurance agreement, insurers increase their share of investments in NAIC 1 private credit by around 2.5 percentage points as opposed to 1.4 percentage points. This is consistent with the fact that in affiliated reinsurance agreements, the insurance group is able to earn the higher yield, whereas in an unaffiliated reinsurance agreement, the insurer can only benefit from improved terms on the reinsurance agreement.

Finally, we apply the same research design to examine total annuity premiums collected. Two years following the execution of a modco or funds withheld reinsurance agreement with a PE-backed Bermuda reinsurer, insurers experience an increase in total premiums of around $820 million one and two years after. These effects are also persistent, although our estimates become noisy in subsequent years. For non-PE reinsurers we again find no effects. These results suggest that insurers jointly determine their insurance product and investment decisions, and are able to utilize offshore reinsurance agreements to grow their annuity businesses. With an affiliated reinsurance agreement, there is a direct incentive of selling more insurance, since the higher-yielding investments increase the profitability of writing insurance. For an unaffiliated reinsurance agreement, a reduction in the cost of writing insurance through improved terms on their reinsurance will also improve the profitability of writing insurance, and lead the insurer to sell more insurance.

Related Literature

First, this paper contributes to the literature on life insurers as regulated financial intermediaries whose product-market and investment decisions are jointly shaped by balance-sheet frictions. Koijen and Yogo (2015) show that financial and statutory-reserve frictions affect the pricing of life insurance and annuity products, while Becker and Ivashina (2015) document reaching for yield by insurance companies in corporate bond markets. Koijen and Yogo (2016) show that captive arrangements ("shadow insurance") allowed life insurers to relax capital requirements through reinsurance arrangements. These papers cover the life insurance industry before 2015 and prior to the captive and reinsurer collateral regulation changes. Our paper shows that following the change in regulatory environment, the role of reinsurance has evolved to include accessing higher-yield investments such as in private credit.

Second, we contribute to the growing literature on private equity activity in life insurance. Kirti and Sarin (2024) study PE acquisitions of life insurers and show that PE owners generate value partly through regulatory and tax arbitrage, with implications for consumer welfare. Kim et al. (2025) show that PE ownership changes insurer asset management, increasing private-placement

investments and credit risk. Meisenzahl et al. (2025) further document the importance of PE-owned insurers in the growth of life insurers' private credit investments and annuity market share. We shift the focus from PE ownership of insurers to PE-backed reinsurers.

Third, the paper contributes to the literature on private credit market. Existing work has emphasized the growth of nonbank lending, the borrower-side determinants for use of private credit, and the potential financial-stability risks stemming from bank and nonbank interconnections (Chernenko et al., 2022; Cai and Haque, 2024; Berrospide et al., 2025). We add to this literature by studying the investor-balance-sheet side of private credit demand for life insurers. Specifically, we demonstrate how private credit growth is affected by insurance regulation and reinsurance and the needs of life insurers, highlighting an important channel alongside traditional borrower-driven demand.

2 Life (re)insurance

2.1 Life Insurance Markets and Insurers

Traditionally, life insurers sold insurance products that provide payments contingent on the death of the insured, allowing households to insure beneficiaries against loss of income or wealth. Today, the term "life insurer" namesake can be misleading, as a larger proportion of business is from annuity products.[5] While traditional life insurance protects households against mortality risk, annuity products shield individuals against longevity or retirement-income/investment risks. Annuity products typically feature two phases: accumulation and drawdown. The drawdown phase – the more traditional view of annuities – allows policyholders to "annuitize" their wealth to protect against longevity risk; the policyholder pays a lump sum premium and in exchange receives a calibrated periodic payments over their remaining lifetime. Conversely, the accumulation phase is effectively an investment vehicle, whereby the policyholder invests premiums with the insurer and receives a credited rate of return that depends on the specific type of annuity product. For example, fixed annuities provide a guaranteed return, while fixed indexed annuities provide a return directly tied to an index with limits on downside and upside. Beyond these business lines, life insurers are also the primary seller of long-term care insurance policies and almost all long-term care insurance is now bundled with a life insurance or annuity product.[6] This is due to long-term care insurance sharing the same long-lived nature of policies as life insurance and annuity products.

Fundamentally, a central feature of these products is that they create long-duration liabilities. Premiums are often collected years or decades before benefits are paid, and insurers must remain

[5] In 2024, 72% of premiums which life insurers received was from annuity policies.

[6] In 2024, 99% of premiums earned for long-term care insurance by life insurers was through bundled products.

solvent over long horizons to honor contractual promises to policyholders. To protect policyholders, life insurers are subject to statutory reserve requirements that require them to hold assets against expected future obligations. These reserves are calculated using actuarial assumptions about mortality, lapses, interest rates, expenses, and other risks. Because reserves must be backed by assets, life insurers are not simply sellers of insurance contracts; they are also large-scale asset managers. The liability structure of their products creates a continuing need to invest premium income and accumulated reserves in assets that can support future policyholder payments. This balance-sheet structure makes life insurers major institutional investors. For example, Federal Reserve Financial Accounts data indicate that life insurers held $3.5 trillion of corporate and foreign bonds in 2024, roughly 23 percent of that asset category across all sectors (Board of Governors of the Federal Reserve System, 2026, Tables L.116 and L.213). Consequently, life insurers' strategic decisions, including insurance product pricing, are shaped as much by financial market regulations, and their incentives as investors as they are by traditional product competition incentives.

A particularly important form of financial regulation is risk-based capital (RBC). Whereas statutory reserves require insurers to hold assets against expected future policyholder obligations, RBC requires insurers to hold capital against the risks created by those obligations and by the assets used to back them. RBC is calculated using a statutory formula and represents the capital the insurer is required to hold. Various regulatory actions can be taken if an insurer's capital is close to or below their RBC.[7] In practice, most life insurers hold capital well above their RBC. The NAIC life RBC formula assigns capital charges to several broad categories of risk, including asset risk, insurance risk, interest-rate risk, and other business risks. As a result, two insurers with the same level of reserves may face different capital requirements depending on the riskiness of their investment portfolios. RBC regulations therefore create a direct link between product-market and investment decisions: writing life insurance or annuity contracts alters an insurer's liabilities, but the assets chosen to support those liabilities also determine the amount of regulatory capital the insurer must hold. In this manner, capital regulation affects not only insurer solvency, but also the relative attractiveness of different products, asset classes, and risk-transfer arrangements.

[7] For example, regulators might require an RBC plan identifying the causes of the capital deficiency, proposed corrective actions, and financial projections. While in more extreme cases, state Commissioners are authorized to place the insurer under regulatory control.

2.2 Life Reinsurance

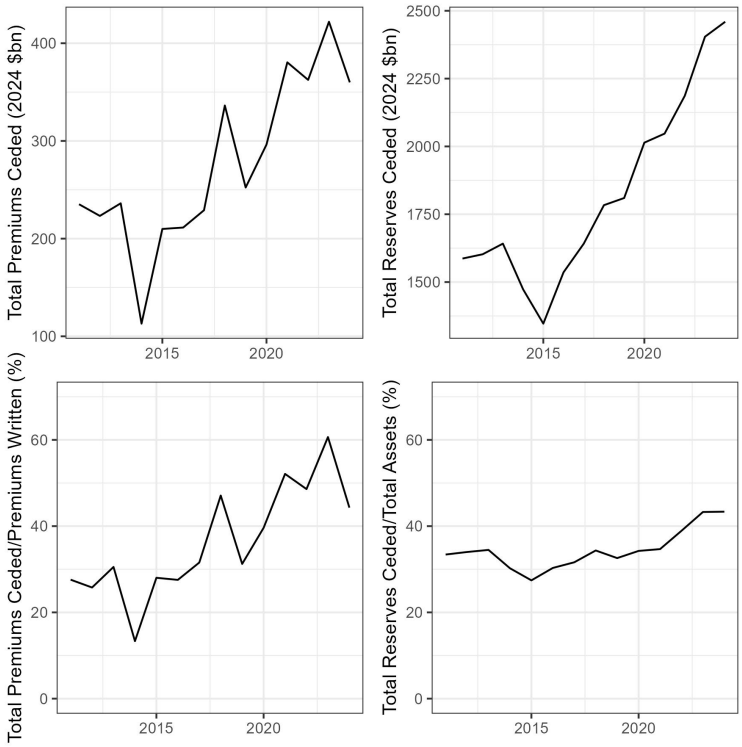


Figure 1: Total Life Reinsurance Ceded (2011-2024)

Reinsurance is often conceptually described as "insurance for insurers". This analogy is particularly apt in health and property & casualty (P&C) contexts, where primary insurers might seek protection against tail risks in medical spending or natural disasters. In contrast, because life mortality risk is highly measurable and diversifiable at a portfolio level, reinsurance serves a slightly different purpose for life insurers.

First, life insurance is an asset-intensive business. Since life insurers are subject to financial frictions (Koijen and Yogo, 2015) when life insurers face balance sheet constraints including from RBC or statutory reserves, they can use reinsurance to shift liabilities, assets and required reserves to a reinsurer, as opposed to directly raising capital. As long as there is a real risk transfer, credit-for-reinsurance rules allow reinsurance to reduce an insurer's RBC. Second, life reinsurance allows insurers to manage their interest rate risks. By ceding blocks of insurance policies to a reinsurer and the underlying assets, they also cede any losses and returns on those assets to the reinsurer.

Total life reinsurance usage has grown dramatically over the past decade. Total reserves ceded reached nearly $2.5tn in 2024 (figure 1). Some of this reflects greater uptake of life reinsurance with total reserves as a fraction of total invested assets reaching 43% in 2024, while some reflects broader growth in the life insurance industry requiring capital relief and interest rate risk protection provided by life reinsurance.

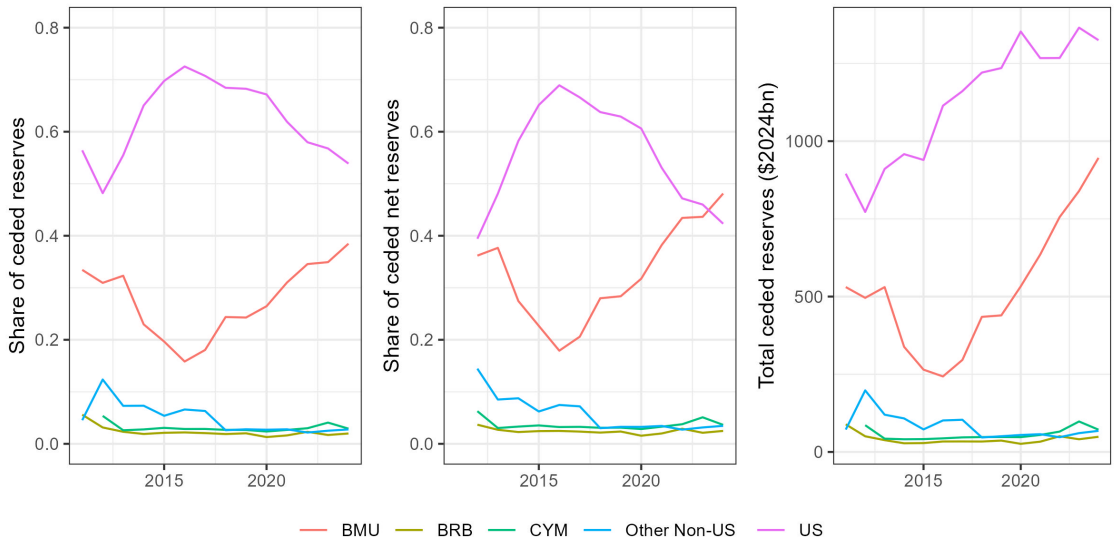


Figure 2: Life Reinsurance Ceded by Domicile (2011-2024)

Reinsurance agreements are characterized by two important dichotomies. First, whether the reinsurer is domiciled in the US or offshore which affects the regulatory regime that largely governs the reinsurer. And second, whether the reinsurance is affiliated or unaffiliated. Although risk is always transferred across *legal* entities in a reinsurance agreement, there is no risk reduction to the broader insurance group with an affiliated reinsurance agreement.[8] As we described before, even without risk transfers outside of the insurance group, affiliated reinsurance can still be important for capital management purposes.

One of the biggest developments in life reinsurance over the past decade has been the growth in offshore reinsurance, in particular insurance ceded to reinsurers domiciled in Bermuda. In the first half of the 2010s, there was a marked drop in insurance ceded to reinsurers in Bermuda due to significant changes in regulation to increase scrutiny and raise the standards of a particular type of affiliated reinsurance – captives (figure 4 and appendix figure 16).[9]

Since the regulatory contraction, however, reinsurance ceded to Bermuda has more than recovered, reaching almost 40% in share of reserves (left) and almost $700bn in 2011 dollars (right). To

[8]Insurers often require reinsurers to secure their obligations with collateral which can mitigate counterparty risk for the ceding legal entity, but this primarily affects the allocation and priority of claims within the insurance group rather than transferring risk outside the group.

[9]Captive reinsurers are affiliated insurers purely for the purposes of reinsuring business written by an operating insurer. In life insurance, they were used extensively to finance reserves for certain term and universal life policies. Regulators were precisely concerned about the extent of true risk transfers in these arrangements, and that some captive obligations were supported by letters of credit, parental guarantees, or other nontraditional collateral rather than standard regulator-approved assets. Following the controversy over such "shadow insurance" arrangements, NAIC introduced reforms which sought to standardize the review of captive arrangements and require sufficient high-quality "primary security" to support ceded reserves.

capture this dynamic more precisely, the middle panel of Figure 2 plots our constructed measure of "net" reserves. This metric accounts for the fact that many US life insurers both assume life insurance from other US life insurers and cede life insurance to offshore reinsurers. In constructing 'net' reserves we remove the amount of life insurance ceded to other US insurers that is then ceded offshore at the insurer level. In other words, from a balance sheet perspective, we determine the 'ultimate' ceding destination of the reinsurance. This balance-sheet adjustment reveals that reinsurers in Bermuda now assume the majority of life insurance by US life insurers (48% in 2024). This paper aims to unpack the reasons for this growth. Reinsurance to other jurisdictions outside of the US play a minor role throughout the data. As a result, this paper focuses its offshore analysis strictly on the Bermuda market.

While unlikely to be a significant driver in and of itself, this growth was partially enabled by a series of regulatory changes. Specifically, the NAIC's reforms to credit-for-reinsurance rules reduced an important regulatory friction facing offshore reinsurers. Historically, a U.S. ceding insurer generally received statutory statement credit for reinsurance ceded to a foreign reinsurer only to the extent the reinsurer's obligations were appropriately secured. Where reinsurance qualifies for credit and transfers relevant risks, it may also reduce associated reserve and RBC requirements. The NAIC's 2011 revisions introduced a framework under which "certified reinsurers" domiciled in approved "qualified jurisdictions," including Bermuda, could satisfy reduced collateral requirements. These were typically adopted by states beginning in 2015. Further revisions in 2019 created "reciprocal jurisdiction reinsurers," allowing qualifying reinsurers domiciled in reciprocal jurisdictions to provide reinsurance to U.S. cedents without posting collateral.

These regulatory changes enabled more reinsurers to operate in the market. In 2015, 46 Bermuda-domiciled reinsurers assumed insurance in unaffiliated agreements. Over 2016-2024, 36 new Bermuda-domiciled reinsurers entered into unaffiliated reinsurance agreements. This also helped reduce the market concentration in the Bermuda reinsurance market for unaffiliated reinsurance: the top 5 share in Bermuda fell from around 92% in 2015 to 81% in 2024 (figure 3).

Regarding the choice between affiliated versus unaffiliated reinsurance, Koijen and Yogo (2016) have previously documented the extensive use of affiliated reinsurance in the first decade of the 2000s to reduce regulatory frictions using other jurisdiction regulatory regimes. One trend worth highlighting is that while affiliated reinsurance continues to be important and the larger slice of the pie, there has still been substantial growth in unaffiliated reinsurance (figure 4). This could reflect the role reinsurance has played in facilitating investment opportunities. Many reinsurers domiciled in Bermuda are well-known asset managers. Through reinsurance, these asset managers can secure investments. Although insurers cannot directly share in the returns from these investments, offshore reinsurers can offer better terms on the reinsurance treaties to attract insurers.[10]

[10]To be precise, regulation does not forbid insurers from sharing in returns from a reinsurance agreement. However,

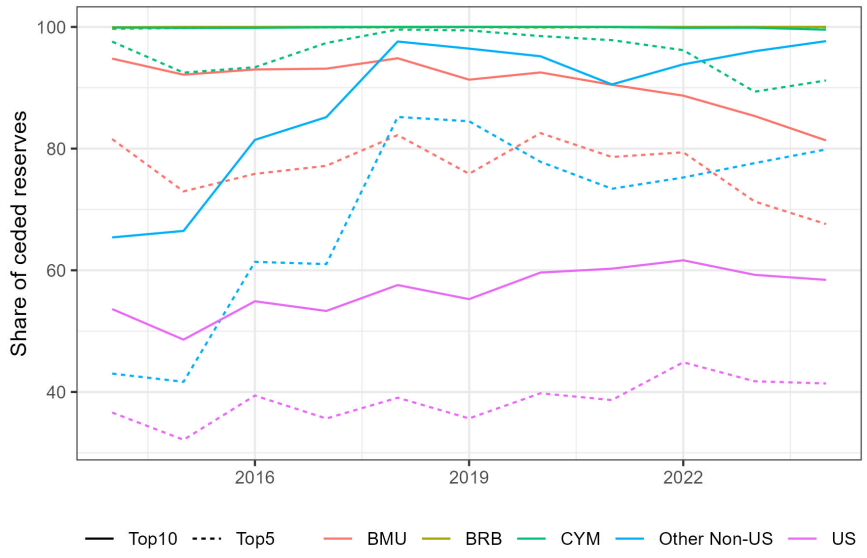


Figure 3: Top 5 and Top 10 shares across jurisdictions for unaffiliated reinsurance

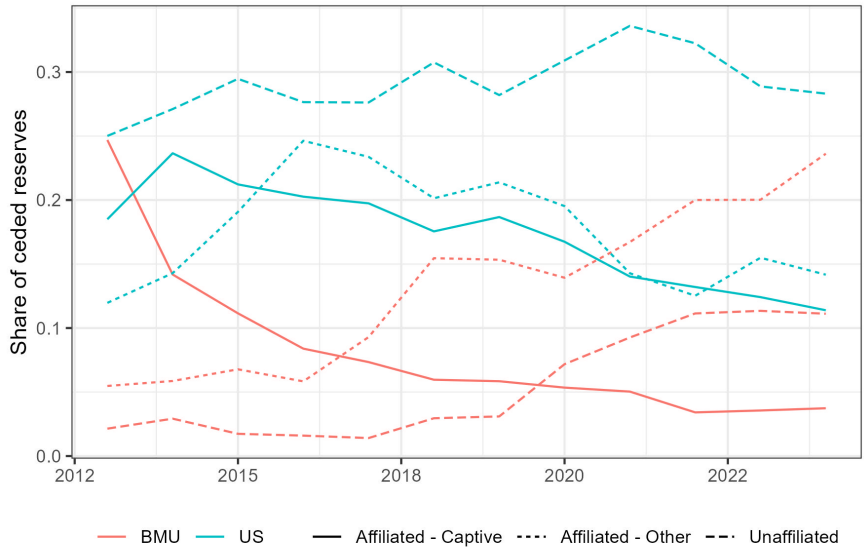


Figure 4: Life Reinsurance Ceded to US and Bermuda by Affiliation

Finally, there are two more important distinctions for how reinsurance agreements work in practice. First, since life reinsurance involves cross-border transactions, exchanging custody of assets can be difficult. Today, most life reinsurance agreements – both affiliated and unaffiliated – are modified coinsurance (modco) or funds-withheld reinsurance agreements (figure 5). In these types of reinsurance, the underlying assets remain on the ceding insurer's balance sheet, even though meaningful insurance and investment risks are transferred to the reinsurer. In a funds-

if the insurer wants to receive regulatory capital relief and reduce their RBC from the reinsurance agreement – a key reason for reinsurance in the first place – then the reinsurer must bear all the interest rate risk.

withheld transaction, the reinsurer assumes the reserves, but the cedent retains the assets and records a corresponding funds-withheld liability to the reinsurer; investment income on those retained assets is then credited to the reinsurer. In a modco transaction, both the assets and the statutory reserves remain on the cedent's balance sheet, with periodic modco reserve adjustments used to allocate the economics of the reinsured block between the cedent and reinsurer.

In summary, while the ceding insurer retains legal ownership of the assets, the true economic exposure is fully transferred to the reinsurer, who typically plays a significant role in portfolio choice, either through contractual investment guidelines or through an affiliated investment manager. Besides reducing transaction costs, there is also another benefit to the ceding insurer by retaining legal ownership of the assets; the ceding insurer has clear visibility over how the assets are invested, reducing moral hazard concerns. This paper focuses on these types of reinsurance, as the fact that the underlying assets stay on the ceding insurer's balance sheet means we can observe how an insurer's investments change with the execution of a reinsurance agreement. Second, life reinsurance agreements are deployed to support particular 'blocks' of insurance and not the insurer as a whole. Much of the growth in reinsurance has been for fixed and indexed annuities (figure 6).

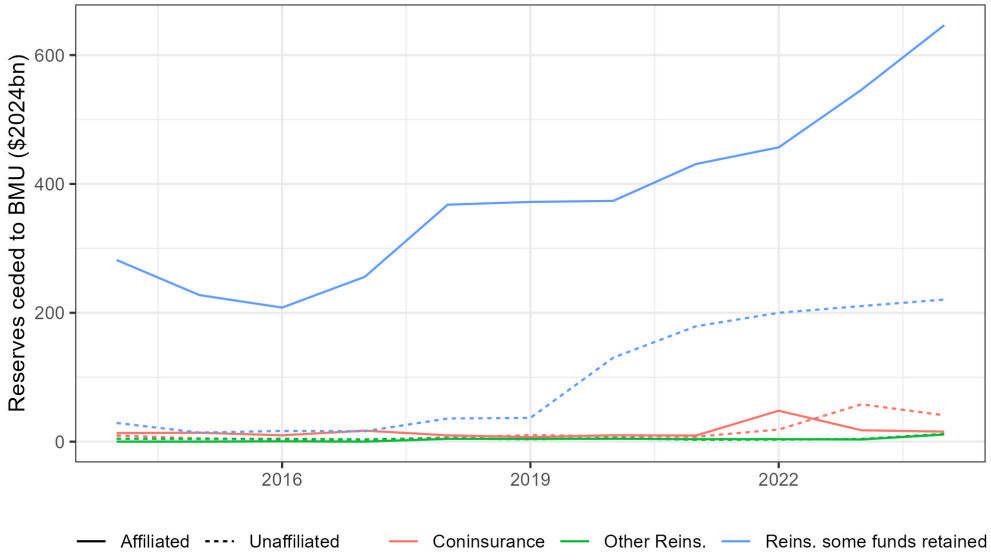


Figure 5: Life Reinsurance Ceded to Bermuda by Affiliation and Type of Reinsurance

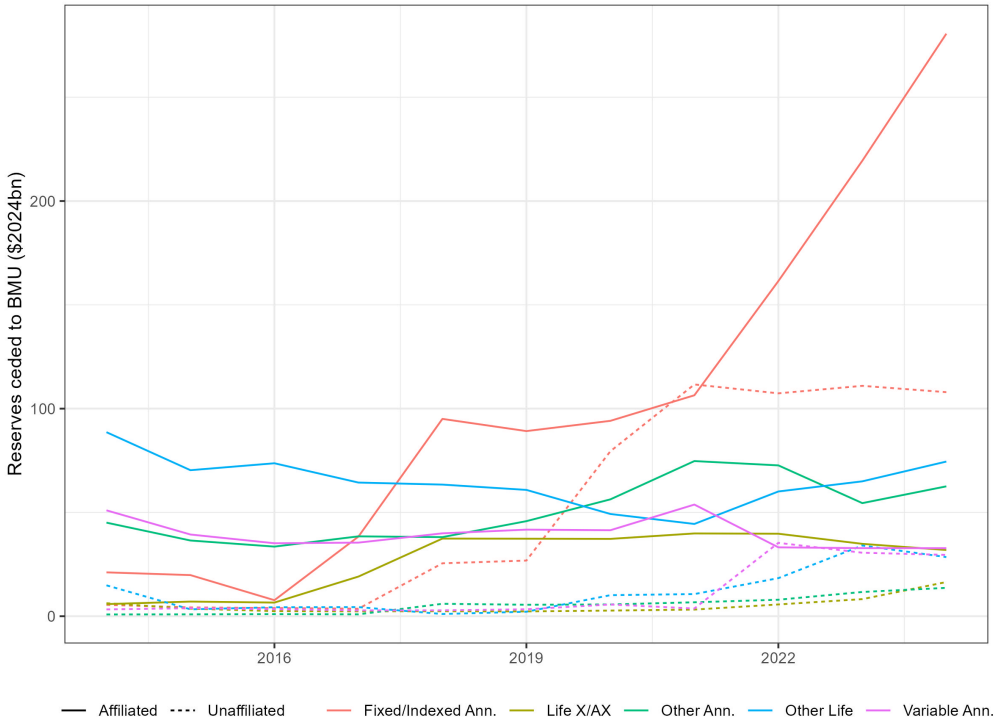


Figure 6: Life Reinsurance Ceded to Bermuda by Affiliation and Business Line

3 Private matters

3.1 Private Credit

Private credit refers to debt financing that is originated or held outside traditional public bond markets and broadly syndicated loan markets. In its narrowest form, the term usually refers to direct loans made by nonbank lenders — such as private credit funds, business development companies, or affiliated asset managers — to middle-market firms. These loans are typically negotiated privately, held to maturity, and tailored to the borrower rather than traded in liquid public markets.

This asset class has grown rapidly over the past fifteen years. Federal Reserve estimates put private credit at $1.34 trillion in the United States and nearly $2 trillion globally by the 2nd quarter of 2024, roughly five times its size in 2009 (Berrospide et al., 2025). The growth of private credit reflects both borrower and investor demand. On the borrower side, private credit offers speed, fewer disclosure requirements, and more customized terms than bank lending or public debt issuance. These features are especially valuable for middle-market firms, private-equity-sponsored firms, and firms that may have difficulty obtaining credit from banks or public markets. For investors, private credit has been attractive because it offers diversification and relatively high yields as compensation for holding illiquid assets. Its expansion has also been supported by broader

structural changes in credit markets: post-crisis bank regulation made some forms of lending less attractive for banks, private equity created demand for flexible debt financing, and large institutional investors, including life insurers, searched for yield during the prolonged low-interest-rate period following the Global Financial Crisis.

However, rapid growth has invited heightened scrutiny from both investors and regulators. Many private credit borrowers are highly levered and the limited trading of private loans and the lack of public information about borrowers can make valuations difficult to assess, especially in periods of stress. For example, in 2025, the bankruptcies of First Brands and Tricolor attracted attention because they revealed opaque financing arrangements and allegations of collateral problems or fraud (Anand et al., 2025). Recent concerns have focused especially on private credit's exposure to software borrowers which JPMorgan estimates to account for roughly one-fifth of private credit exposure, rising substantially when broader technology and business-services exposure is included (J.P. Morgan Private Bank, 2026). Many direct lenders expanded aggressively into software and software as a service firms because recurring revenues appeared stable and scalable, but some loans were underwritten against annual recurring revenue rather than current profitability, leaving borrowers vulnerable to higher interest rates, slower growth, and potential AI-driven disruption of software business models.

Although private credit is discussed very frequently, the boundaries of the asset class are not precise: some definitions focus only on direct corporate lending, while broader definitions also include asset-based finance, private placements, infrastructure debt, real estate debt, or other privately originated credit. In this paper, we consider private credit as insurer holdings of privately placed or privately rated corporate debt, direct loan assets, and credit-oriented private fund, partnership, or joint venture exposures, including vehicles backed by bonds or mortgages, while excluding vehicles that appear to be hedge funds or private equity (see section 4.1 for the specific coding rules). Under our definition of private credit, in aggregate life insurers held around $440 billion in private credit or 7.8% of their invested assets in 2024 (figure 7). The vast majority of this is in schedule D part 1 assets – private placements or privately rated corporate debt securities – and 46% of this is NAIC 1 designated.

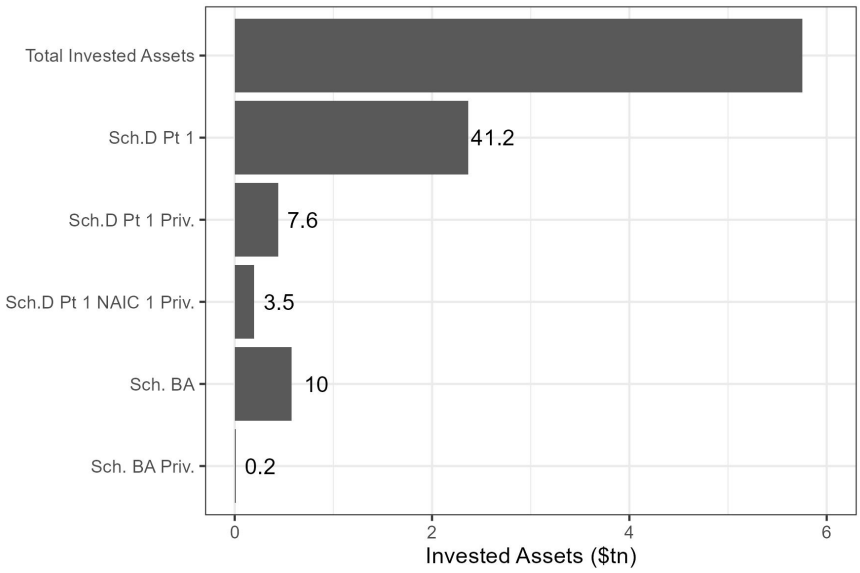


Figure 7: Life Insurer Invested Assets in 2024

In aggregate, life insurers hold a large proportion of all private credit investments, but a smaller group of life insurers hold the majority of private credit investments (figure 8). For instance in 2024, the median share in schedule D part 1 private credit was around 3.9% and 75% of insurers had investments in schedule D part 1 private credit totaling to $0.68 bn.

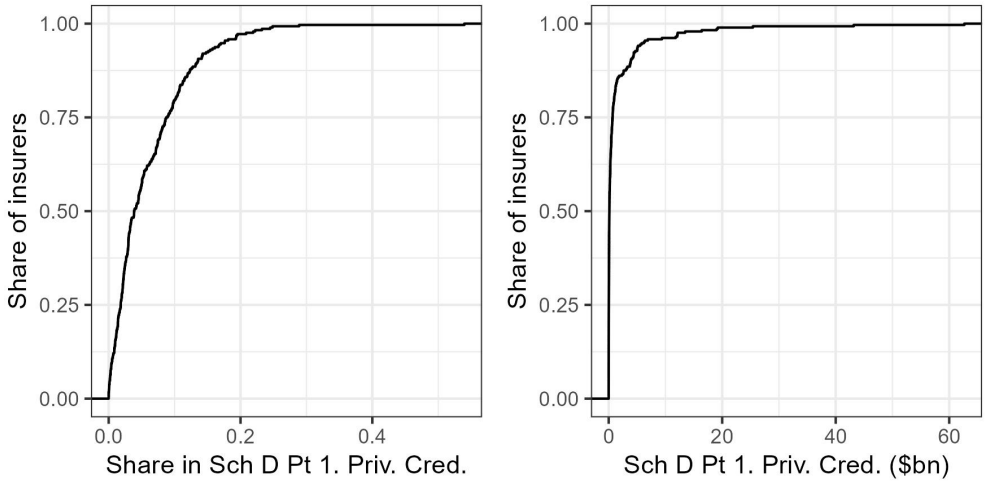


Figure 8: Distribution of Private Credit Among Life Insurers in 2024

3.2 Private Equity

PE ownership of life insurers has received considerable scrutiny in recent years. From 2005 to 2023 there have been at least 129 acquisitions by PE (Kirti and Sarin, 2024; Kim et al., 2025; Johnson and Carelus, 2024). From an asset-liability perspective, this transaction structure offers clear synergies. Life insurers generate large, stable pools of long-duration liabilities, while private equity managers

specialize in originating and managing higher-yielding illiquid assets.

In addition to the rapid increase in the number of PE-owned insurers, a growing body of work finds that PE-owned insurers differ in ways that could be of concern to regulators. PE-owned life insurers increase their share of investments in more illiquid private placements (Meisenzahl et al., 2025), in lower investment-grade private placements without increasing diversification across bond issuers (Kim et al., 2025) and engage more in regulatory arbitrage (Kirti and Sarin, 2024). Nevertheless, PE-owned insurers play a relatively small role in the overall life-insurance industry, including in their investments in private credit (figure 9). At a portfolio level, however, the average PE-owned insurer invests more in private credit with an average share of 9.7% versus 5.5% for non PE-owned insurers.

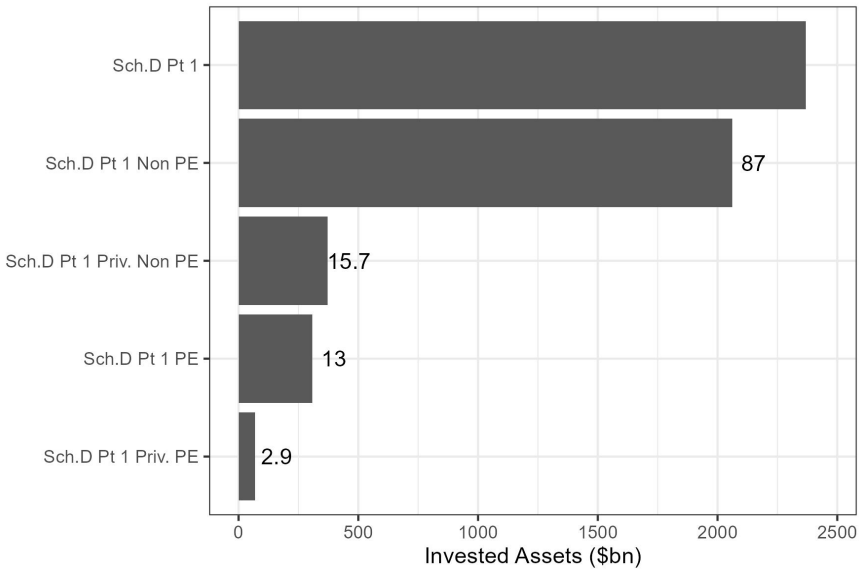


Figure 9: PE vs Non PE Life Insurers' investments in Sch D pt 1 in 2024

In this paper, we shift our focus to the role of PE-backed reinsurers. While PE-backed primary life insurers are not the majority of life insurers, PE-backed life reinsurers are the majority by both count and share of reserves assumed (table 1). The significance of PE in reinsurance is unsurprising. PE may prefer accessing life insurer assets through reinsurance rather than acquisitions as it allows them to abstract from the administration of insurance policies and undesired blocks of insurance (eg: blocks in health insurance) and focus on managing large pools of assets.

Table 1: PE in Modco/FWH BMU Reinsurance

	PE-Backed Insurers	PE-Backed Reinsurers
Number of Firms	11	40
Share of Firms (%)	13	62
Reserves Ceded/Assumed ($bn)	259	580
Share of Reserves Ceded/Assumed (%)	30	68

Note: All PE-Backed Insurers contract with PE-Backed Reinsurers and almost all of their reserves ceded are to PE-Backed Reinsurers.

4 Data

NAIC Data:

We draw on a wealth of data from life insurer annual statutory financial statements filed with the NAIC to construct panels for life insurance companies from 2011-2024 about their investments and executed reinsurance agreements. Schedule D, Part 1 provides investment-level information about long-term bonds held by insurers on Dec 31 of a given year. This includes information about their NAIC designation, book adjusted carrying value and other information for identifying what type of bond it is. Schedule BA, Part 1, provides similar information for other invested long-term assets. As discussed previously, we define private credit as insurer holdings of privately placed or privately rated corporate debt, direct loan assets, and credit-oriented private fund, partnership, or joint venture exposures, including vehicles backed by bonds or mortgages, while excluding vehicles that appear to be hedge funds or private equity. More specifically we tag the following investments as private credit:

- From Schedule D, part 1, any investment:
  - with a CUSIP with any of *, @ or # in any position from 6-8, *OR*,
  - with a Corporate 'bond' line number label (see Appendix A and National Association of Insurance Commissioners (2024)) *AND* having a SVO administrative symbol that indicates a Private Letter Rating.
- From Schedule BA, part 1, any investment:
  - with Collateral loans *OR* Non-collateral loans in the line number label (see National Association of Insurance Commissioners (2024)), *OR*,

- Is a 'Non-registered Private Fund with Underlying Assets Having the Characteristics of' *OR* 'Joint Venture, Partnership or Limited Liability Company Interests with Underlying Assets Having the Characteristics of' investment according to line number labels, *AND*
  - Has one of 'Bonds' *OR* 'Mortgage' in the line number label name, *AND*
  - Does not have a hedge fund *OR* private equity, 'Type and Strategy'.

Schedule S, Part 3, Section 1 provides information about reinsurance agreements on blocks of life insurance (including annuity policies), when the life insurer is ceding insurance and domiciled in the US. This data includes information about the reinsuring counterparty (name, domicile and any relevant identifiers), the type of reinsurance agreement, date of execution, annual premiums ceded and reserve credit taken (representing reserves transferred to the reinsurer). Note that we do not directly observe the treaty-specific asset pool for each agreement.[11] In addition to Schedule S and D, we also draw upon data from the assets statement, Exhibit 1 for premiums written and NAIC's list of PE-backed insurers maintained by the Capital Markets Bureau using schedule Y.

PE-backed Reinsurers Domiciled in Bermuda

We focus on the role of PE in reinsurance market for reinsurers domiciled in Bermuda as this is where the majority of reinsurance activity occurs. However, because these reinsurers are domiciled in Bermuda, NAIC annual statutory financial filings provide relatively little information about them, including whether they are PE backed or not. The Schedule S reinsurance schedules of U.S. life insurers identify Bermuda counterparties by name and identification number, but do not disclose their ownership or affiliation with asset managers. To overcome this limitation, we construct this information by hand.

We begin with the universe of Bermuda-domiciled reinsurers active in the U.S. life reinsurance market, defined as reinsurers that have assumed any reinsurance agreement in NAIC Schedule S during our sample period. We identify 217 unique reinsurers (by NAIC company identification number, or by name for those without an identification number). For each reinsurer, we conduct a structured desktop search to determine whether the entity is owned by, or otherwise affiliated with, a PE or alternative asset management firm. We draw on a range of sources to verify such linkages, including regulatory and corporate disclosures (SEC EDGAR filings, Bermuda Monetary Authority materials, and company filings), specialized reinsurance and insurance trade press (AM Best, Business Insurance, Insurance Journal, and Reinsurance News), PE-firm and reinsurer press

[11] Beginning with the 2025 annual financial filings, insurers are required to report in aggregate, across all agreements, and types of assets, the amount of assets supporting modco and funds withheld reinsurance agreements. However, since we identify private credit investments using investment-level characteristics, these data do not help improve our research design.

releases announcing transactions, and major financial news outlets (Reuters and Business Wire).

We classify a reinsurer as PE-backed if we find a source documenting any of the following relationships between the reinsurer and a PE firm: i) full PE ownership (through acquisition or the PE firm organically establishing a reinsurance company), ii) partial ownership or a minority stake, or iii) a strategic partnership or investment-management arrangement. If we find no source documenting any such association, the reinsurer is considered not PE-backed. This yields 83 PE-backed reinsurers and 134 non-PE affiliated reinsurers. Among the PE-backed reinsurers, the large majority reflect direct ownership (e.g., Apollo–Athene, KKR–Global Atlantic), with a smaller set reflecting strategic partnerships or asset-management mandates rather than control.

5 The Impact of Executing an Offshore Reinsurance Agreement

5.1 Empirical Strategy

To investigate the impacts of reinsurance with a reinsurer domiciled in Bermuda on an insurer's investments and their pricing of insurance products, we leverage the execution of modco or funds withheld (FWH) reinsurance agreements in a staggered difference-in-difference research design. The treatment group consists of insurers who execute a modco or FWH reinsurance agreement with a reinsurer domiciled in Bermuda. More specifically, we focus on the execution of the *first* such reinsurance agreement in the period 2015-2024. The beginning of this period reflects the beginning of when certified reinsurers were introduced and the tightening of captive regulations which significantly affected the nature of offshore reinsurance. We also focus on the 'first' instance of such an agreement. This reflects that over time insurers often execute multiple reinsurance agreements, sometimes with the same reinsurer or with other reinsurers.[12]

We also remove from treatment (and control) any insurers who i.) execute a reinsurance agreement in any other offshore jurisdiction before the first such reinsurance agreement with a reinsurer domiciled in Bermuda or ii.) execute a reinsurance agreement with a reinsurer domiciled in Bermuda which is not a modco or FWH reinsurance agreement before a modco or FWH reinsurance agreement. The first case is because we are focused on identifying whether each reinsurer in Bermuda is PE-backed or not. The second case is because the assets backing the reinsurance agreement are moved to the reinsurer's balance sheet in non modco or FWH reinsurance agreements meaning we cannot properly observe how investments change.

[12] An alternative research design we could have adopted is to use a stacked difference-in-difference design leveraging each adoption of a reinsurance agreement. We did not opt for this approach because reinsurance agreements can be executed on arbitrary dates. Thus, at the yearly level, it is difficult to cleanly track pre and post-periods for each reinsurance agreement.

We also vary the treatment groups, to focus on specific comparisons of interest. As the paper focuses on the role of PE-backed reinsurers, we define treatments where the reinsurer is PE-backed and where the reinsurer is not PE-backed. Another distinction of interest is whether the reinsurer is affiliated or not. Under an affiliated agreement, investment risks and gains accrue to the insurer group and there is no risk transfer (to the broader insurance group) unlike with an unaffiliated agreement. Therefore we might expect more pronounced changes on investment portfolios with an affiliated agreement compared to an unaffiliated agreement. Table 2 summarizes the number of insurers included in treatment groups by execution year, whether the reinsurer is PE-backed, and for those which are PE-backed whether they are affiliated or not. Most insurers in the treatment group are those who contract with a PE-backed reinsurer. Of those who contract with a PE-reinsurer the split between affiliated versus unaffiliated agreements is even. We do not study affiliated versus unaffiliated reinsurance when the reinsurer is not PE-backed, simply because there are not many non PE-backed reinsurers in the treatment to begin with.

The control group consists of units not yet treated and insurers who never execute such a reinsurance agreement in the sample. In addition to the above exclusions from control, we also drop any insurers who never execute a modco or FWH reinsurance agreement with an offshore reinsurer, but do execute a different type of reinsurance agreement with an offshore reinsurer. Our never-treated control group consists of 181 insurers.

Table 2: Treatment Group

Executing Year	With a non-PE reinsurer	With a PE reinsurer	With an affiliated PE reinsurer	With a non-affiliated PE reinsurer
2015	2	2	0	2
2016	0	1	1	0
2017	0	7	7	0
2018	1	7	3	4
2019	0	3	0	3
2020	2	15	5	10
2021	0	10	5	5
2022	3	5	3	2
2023	2	3	1	2
2024	4	3	3	0
Total	14	56	28	28

Our main specification then is the following:

$$\mathrm{Y}_{it} = \alpha_i + \lambda_t + \sum_{\ell \neq -1} \beta_\ell \mathbf{1}\left\{t - E_i = \ell\right\} + \varepsilon_{it}, \tag{1}$$

Where Y_{it} is the share of investments in private credit (or subset of private credit investments), α_i and λ_t are insurer and year fixed effects and β_ℓ captures the treatment effect at time ℓ relative to the execution of the reinsurance agreement.

5.2 Results

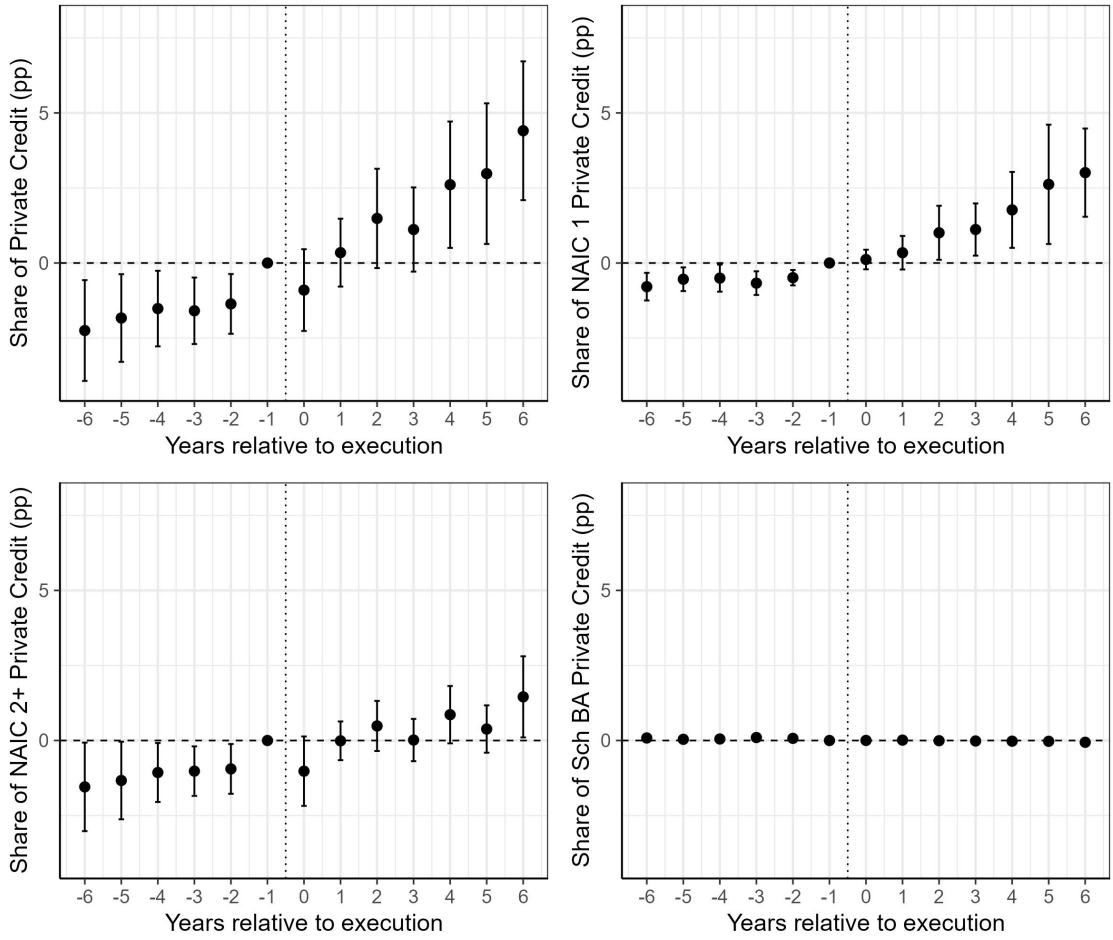


Figure 10: Private Credit Difference-in-Difference Event Studies

Figure 10 plots our staggered difference-in-difference estimates of β_ℓ across overall shares of investments in private credit and sub-categories of private credit investments. Following the execution of a modco or FWH reinsurance agreement with a reinsurer domiciled in Bermuda, insurers appear to increase their share of investments in private credit, particularly those in NAIC 1 designated

long-term bonds. These estimates are relatively noisy however, and we show next that the noisiness is driven by the difference between whether the reinsurer is PE backed or not. Still, the fact that the bulk of the effect is coming from NAIC 1 designated long-term bonds is consistent with how insurers, reinsurers and their affiliated asset managers often describe their private credit investments as 'investment-grade'.[13]

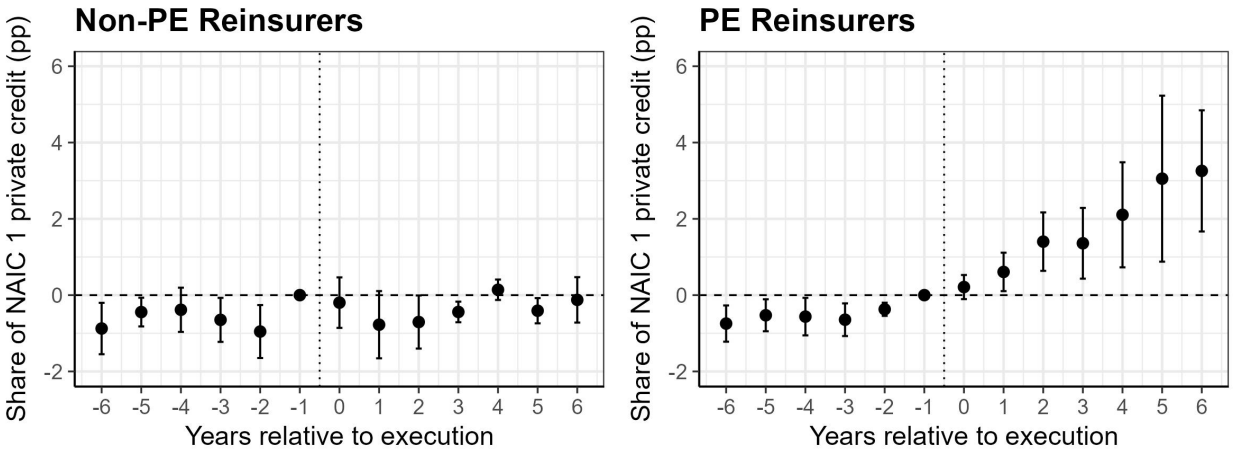


Figure 11: NAIC 1 Private Credit Difference-in-Difference Event Studies

In figure 11 we focus on private credit in NAIC 1 designated long-term bonds and estimate equation 1 separately across cases where the reinsurer is a PE-backed reinsurer (right) versus not (left). It is clear that the noisiness of the previous figure is likely attributed to the non-response of the share of investments in NAIC 1 private credit for when the reinsurer is not PE-backed. In contrast, we observe a sharp increase in the share of NAIC 1 private credit when the reinsurer is PE-backed. 2 years following the execution of the reinsurance agreement, insurers increase their share of investments in NAIC 1 private credit by about 1.40 percentage points. These effects are persistent and growing: 5 years from execution of the reinsurance agreement, the effect grows to around 3 percentage points. The magnitude of our estimates are material in the context of the average and median life insurer which has respectively, 6% and 3.9% of their investments in private credit (in 2024).

Another way of understanding the significance of the result is by contrasting these effects with the effects of PE acquisition of a life insurer. Meisenzahl et al. (2025) finds that two years following a PE-takeover of a life insurer, they increase their share of investments in private placements by almost 6 percentage points. Kim et al. (2025) find an average effect of around 6.37 percentage points. On a per-insurer basis these effects are much larger than what we find. However, from an

[13]For example, see Apollo Global Management's discussion of private credit here (Apollo owns Athene, a major life insurer) and this factsheet by the American Council of Life Insurers.

overall industry perspective, the aggregate effects from PE acquisitions are at most around the size of the aggregate effects from our offshore reinsurance channel. our treated insurers who reinsure with a PE-reinsurer make up around $1.6 trillion in assets in 2024. An estimate of 1.4 percentage points yields an increase of $22.2bn in NAIC 1 long-term bonds in private credit. Kim et al. (2025) identify 25 PE-acquisitions from 2014-2018 with assets totalling around $84bn, thus their estimate of 6.37 percentage points would imply an increase of around $5bn in private credit investments. Meisenzahl et al. (2025) identify 10 PE-acquisitions from 2018-2023, but do not provide enough information to determine the assets represented by those 10. A total of around $350bn in acquired PE-assets would be required to match our aggregate effect size.

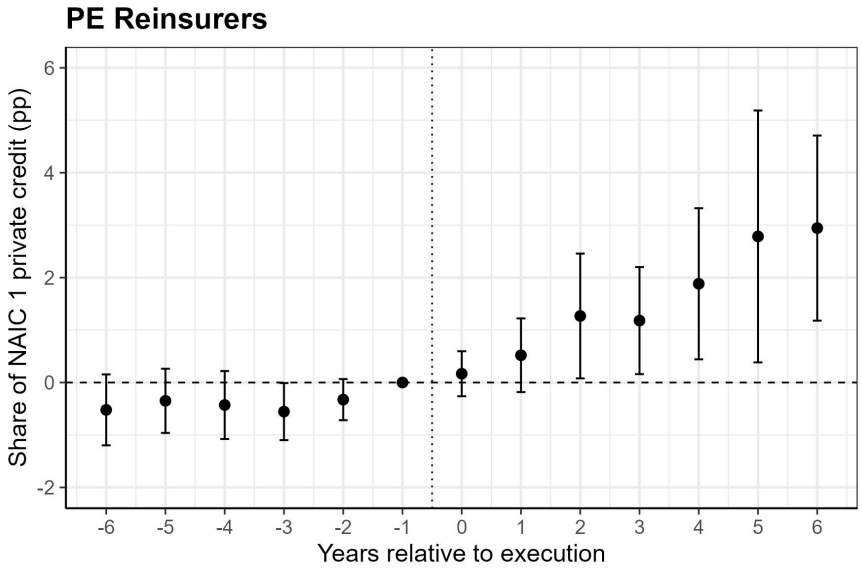


Figure 12: Detrended Difference-in-Difference Event Study

To identify treatment effects in a difference-in-difference research design, the parallel trends assumption needs to be satisfied. That is, we require that outcomes for the treated insurer would've evolved in the same way as the control group if they did not execute a reinsurance agreement. The right panel of figure 11 suggests that there are likely to be pre-trends - the share of investments in NAIC 1 private credit rises slightly before the execution of the reinsurance agreement. Unlike the usual setting, we expect insurers to utilise reinsurance agreements as part of a broader growth strategy. Accessing higher-yield investments increases their economic margins on writing new insurance policies and allows them to lower insurance prices for consumers to expand their market share. Reinsurance is an essential part of this strategy. Reinsurance is required so that insurers can shift capital requirements to the reinsurer, freeing up capital to write more policies. Still, some steps to higher-yield investments can be taken without reinsurance. Therefore, it's reasonable to expect the potential for pre-trends. On the other hand, the relative size of the growth in the treat-

ment compared to the pre-trend suggests that most of the shift into NAIC 1 private credit can be attributed to the execution of the reinsurance agreement. To formalize this, in figure 12 we construct a two-step staggered difference-in-difference estimator which allows for a linear pre-trend. Specifically, in the first step we estimate using never-treated insurers and insurers in the treatment, but before they are treated:

$$Y_{it} = \alpha_i + \lambda_t + \gamma(t - E_i)\delta_i + \varepsilon_{it}, \tag{2}$$

Where δ_i is an indicator for whether i is eventually treated. In short, this step estimates a linear pre-trend $\hat{\gamma}$. Then in the second step we estimate:

$$\tilde{Y}_{it} = \alpha_i + \lambda_t + \sum_{\ell \neq -1} \beta_\ell \mathbf{1}\{t - E_i = \ell\} + \varepsilon_{it}, \tag{3}$$

$$\tilde{Y}_{it} = Y_{it} - \hat{\gamma}(t - E_i)\delta_i$$

In other words, we de-trend Y_{it} using our estimate $\hat{\gamma}$ to obtain $\tilde{Y}_{it}$ and then estimate the same staggered difference-in-difference specification as in equation 1 using $\tilde{Y}_{it}$.[14] This isolates the effect of the reinsurance agreement. We find that the pre-trend is weakened and treatment effects largely remain. For example, 2 years following the execution of the reinsurance agreement, insurers increase their share of investments in NAIC 1 private credit by about 1.27 percentage points (previously 1.40 percentage points).

Given that insurers likely select into executing reinsurance agreements, we might also worry that the characteristics of treated and control insurers are quite different. Table 3 confirms this. We compute averages of characteristics in 2015, across insurers which are treated at some point in our sample versus those which are never treated. Treated insurers are significantly larger, write more annuity policies, invest more in private credit and have slightly lower RBC ratios. Our main result is robust to pre-trends, however concerns could still remain about our comparisons. Accordingly, we construct a matched stacked difference-in-difference estimator. For each treated cohort of insurers defined by the year of first executing a modco or FWH reinsurance agreement with a PE reinsurer domiciled in Bermuda, we construct a matched control group using logistic regression propensity scores. We match the control group using the average of the characteristics from table 3 for the 3 years prior to executing the reinsurance agreement. We stack the matched samples constructed in each cohort to form the estimation sample and re-estimate equation 1.

[14] Correct inference requires accounting for the uncertainty arising from estimating γ. We compute standard errors using a block bootstrap, drawing insurers, over 400 iterations

Table 3: Balance in 2015 by averages

	Total assets (2011 $bn)	RBC	Annuity Reserve Share (%)	Share of Private Credit Investments (%)
Ever Treated	29.0	10.69	68.5	4.7
Never Treated	8.2	12.80	37.4	4.2

The results are presented in figure 13.[15] Since we construct a stacked panel for this estimator we restrict the time window to the 4 years before and 3 years including and after the treated year. Other than that, our results are quite similar with those from figure 11. 2 years following the execution of a reinsurance agreement we obtain a point estimate of an increase in the share of NAIC 1 private credit of 1.43 percentage points.

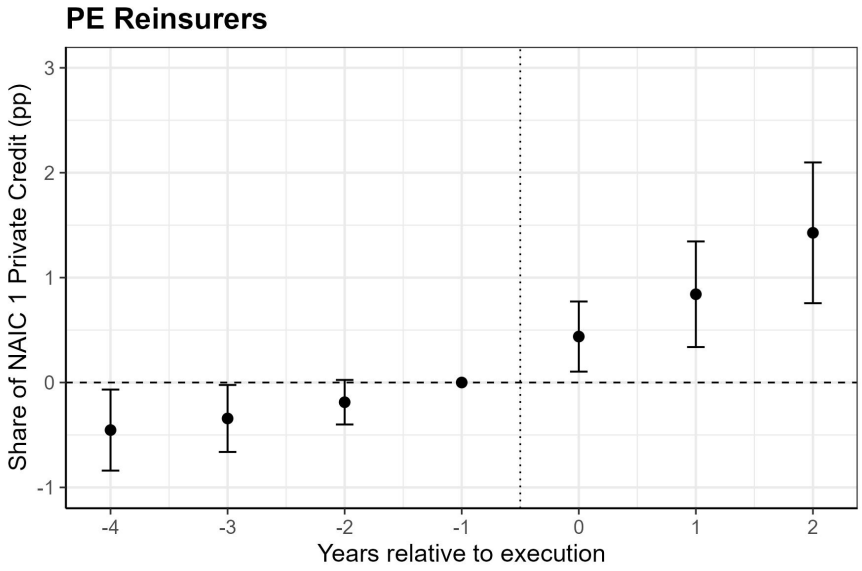


Figure 13: NAIC 1 Private Credit Matched Stacked Difference-in-Difference Event Study

Finally, in figure 14 we explore one important dimension of heterogeneity - whether the PE reinsurer is affiliated or not. To do this, we estimate separately equation 1 across cases where the PE reinsurer is affiliated (left) or not (right). The event study plots suggest that the clearest effects are with affiliated PE reinsurers. In fact, the effect after 2 years increases to almost 2.5 percentage points (compared to 1.4 percentage points), reaching almost 5 percentage points 5 years out. For unaffiliated PE reinsurers, we find that there are still likely effects but somewhat noisy and smaller in magnitude. For example, we do not detect an effect 2 years following, however, 1 year after we

[15] Correct inference requires accounting for the uncertainty arising from constructing the matched control group. We compute standard errors using a block bootstrap, drawing insurers, over 400 iterations

find an increase in the share of investments of around 0.73 percentage points or around 18% of the median share of investments in private credit. These results seem consistent with the fact that for affiliated agreements, the insurer group directly benefits from changes in the insurer's investment portfolio leading to stronger incentives to adjust their investment portfolio. For unaffiliated agreements the insurer might indirectly benefit from better overall terms of a reinsurance agreement but cannot directly share in the investment returns of the reinsurance agreement.

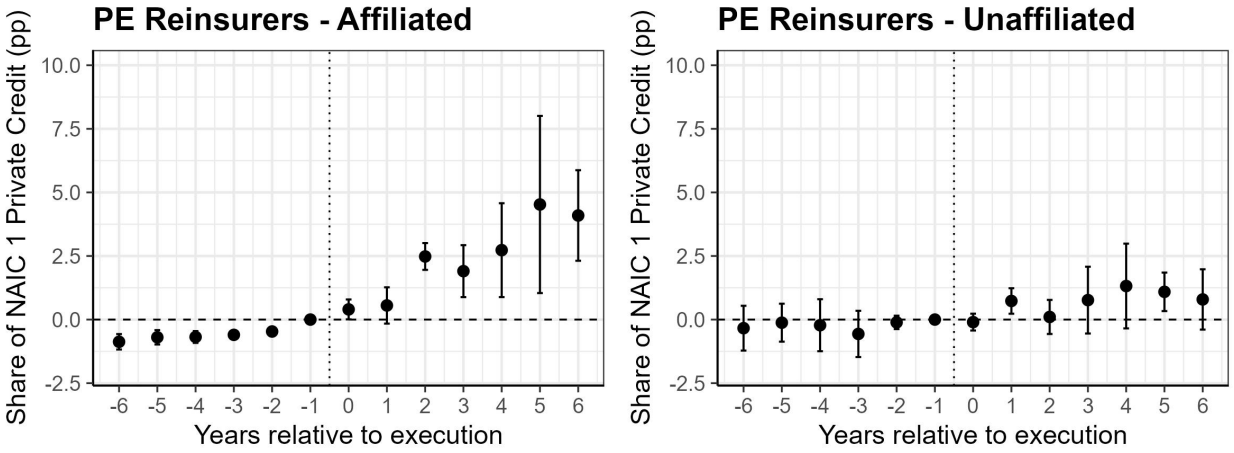


Figure 14: NAIC 1 Private Credit Aff/Unaff Difference-in-Difference Event Studies

5.3 The Impact of Executing an Offshore Reinsurance Agreement in Annuity Markets

Reinsurance affects the profitability of writing insurance. First in general, by relaxing insurer capital constraints, reinsurance can reduce costs. Focusing on the comparisons we make between offshore and domestic reinsurance there are also implications. In unaffiliated reinsurance, insurers can benefit through better terms, while in affiliated reinsurance, the insurance group improves profitability through higher-yielding private credit investments. Thus, we expect the execution of offshore reinsurance agreements to have impacts on the annuity market as insurers adjust their prices.

Drawing on data for total annuity written premiums, We utilize the same staggered difference-in-difference research design to study the impacts of the first execution of a modco or FWH reinsurance agreement with a reinsurer domiciled in Bermuda. We find similar results as with the share of private credit investments. Following the execution of such a reinsurance agreement with a PE-backed reinsurer, insurers increase their total annuity premiums. Specifically, 2 years following execution, insurers earn around $820 million more in annuity premiums. These effects are quite large at around 38% of average total annuity premiums. Again, for non-PE backed reinsurers we detect no effects. There is some evidence that these effects persist, suggesting that insurers

contracting with PE-backed reinsurers are able to leverage the benefits of the offshore reinsurance agreements to expand their annuity business.

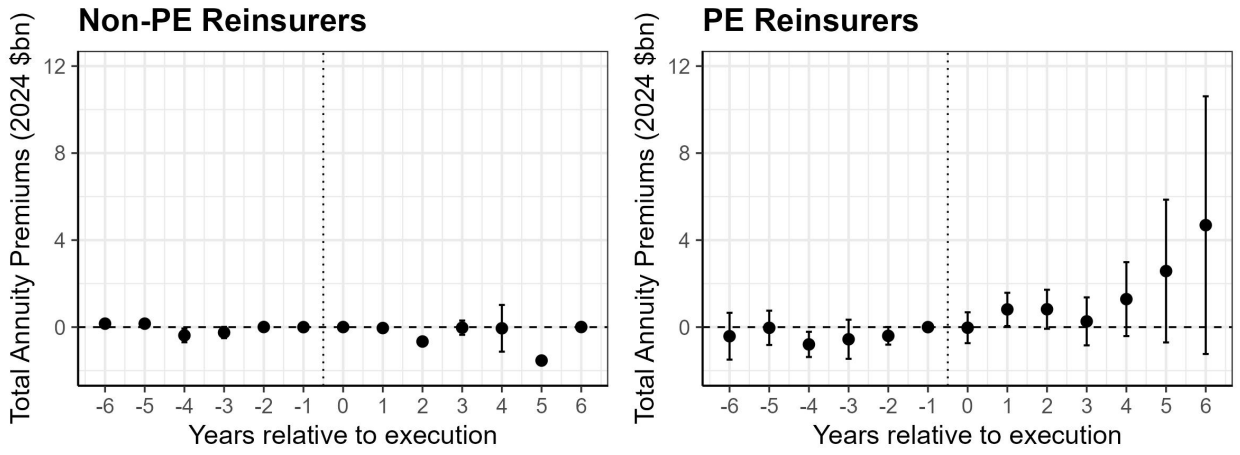


Figure 15: Annuity Premiums Difference-in-Difference Event Studies

6 Conclusion

Following losses and withdrawal caps in 2026, private credit has become an area receiving significant scrutiny by investors and regulators. Since private credit markets have meaningful exposure to software and other recurring-revenue borrowers—where falling valuations, AI-related disruption, and recent restructurings have challenged assumptions about stable cash flows—these concerns are likely to persist. They may also broaden as private credit expands further into consumer credit, including buy-now-pay-later and other unsecured installment lending, where borrower leverage, thin credit histories, and limited transparency make losses harder to assess before a downturn.

While the significant involvement in private credit by life insurers, and PE-backed life insurers is well known, our paper finds that PE-backed reinsurers might be a larger driver for life insurers' involvement in private credit, especially as it provides a channel for non-PE life insurers to invest in private credit. And, while broad private credit concerns are likely to persist, our results also show that life insurer's investments in private credit are likely to persist. Life insurers which take advantage of these offshore reinsurance agreements, are able to expand their annuity business, meaning that this strategy is sustainable in normal times.

Our results also show that simply studying a life insurer's investments in private credit in isolation could be somewhat misleading. Much of it is tied up in modco or FWH reinsurance agreements which means that the underlying risks are borne elsewhere, especially for an unaffiliated reinsurance agreement. Therefore, this paper suggests value in regulators and analysts applying additional scrutiny to the activities of PE-backed reinsurers.

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Appendix

A Private Credit Line Numbers

Table 4: Schedule D Private Credit Line Numbers

Group	Type	2011-2018 Code	2019-2021 Code	2022 -2024 Code
Industrial and Miscellaneous (Unaffiliated)	Issuer Obligations	3299999	3299999	1019999999
Industrial and Miscellaneous (Unaffiliated)	Residential Mortgage-Backed Securities	3399999	3399999	1029999999
Industrial and Miscellaneous (Unaffiliated)	Commercial Mortgage-Backed Securities	3499999	3499999	1039999999
Industrial and Miscellaneous (Unaffiliated)	Other Loan-Backed and Structured Securities	3599999	3599999	1049999999
Parent, Subsidiaries and Affiliates	Issuer Obligations	4999999	4999999	1419999999
Parent, Subsidiaries and Affiliates	Residential Mortgage-Backed Securities	5099999	5099999	1429999999
Parent, Subsidiaries and Affiliates	Commercial Mortgage-Backed Securities	5199999	5199999	1439999999
Parent, Subsidiaries and Affiliates	Other Loan-Backed and Structured Securities	5299999	5299999	1449999999
Parent, Subsidiaries and Affiliates	Affiliated Bank Loans - Issued		5399999	1459999999
Parent, Subsidiaries and Affiliates	Affiliated Bank Loans - Acquired		5499999	1469999999
Unaffiliated Bank Loans	Unaffiliated Bank Loans - Issued		6399999	1819999999
Unaffiliated Bank Loans	Unaffiliated Bank Loans - Acquired		6499999	1829999999

B Other descriptive figures

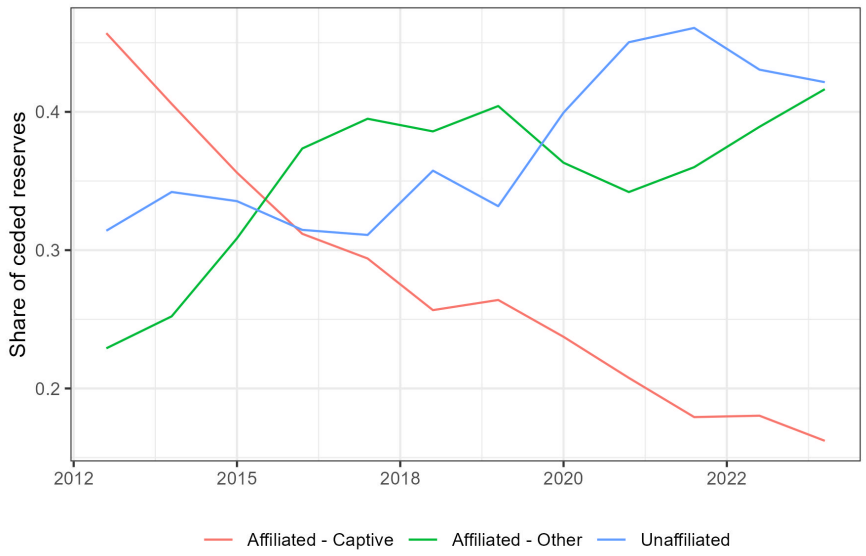


Figure 16: Shares of Life Reinsurance by Affiliation Type (2013-2024)